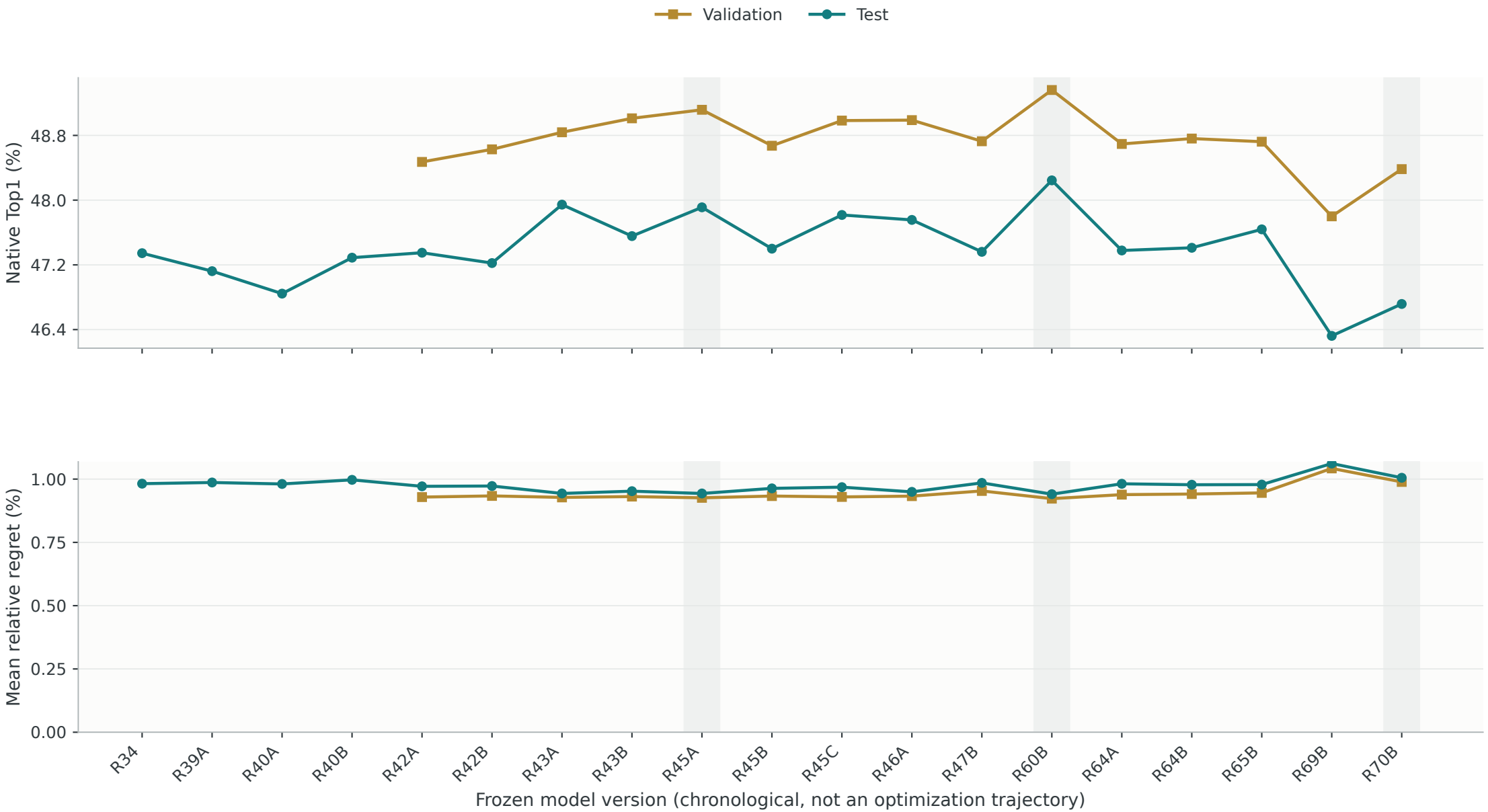
Frozen versions: full-18 validation and test trends

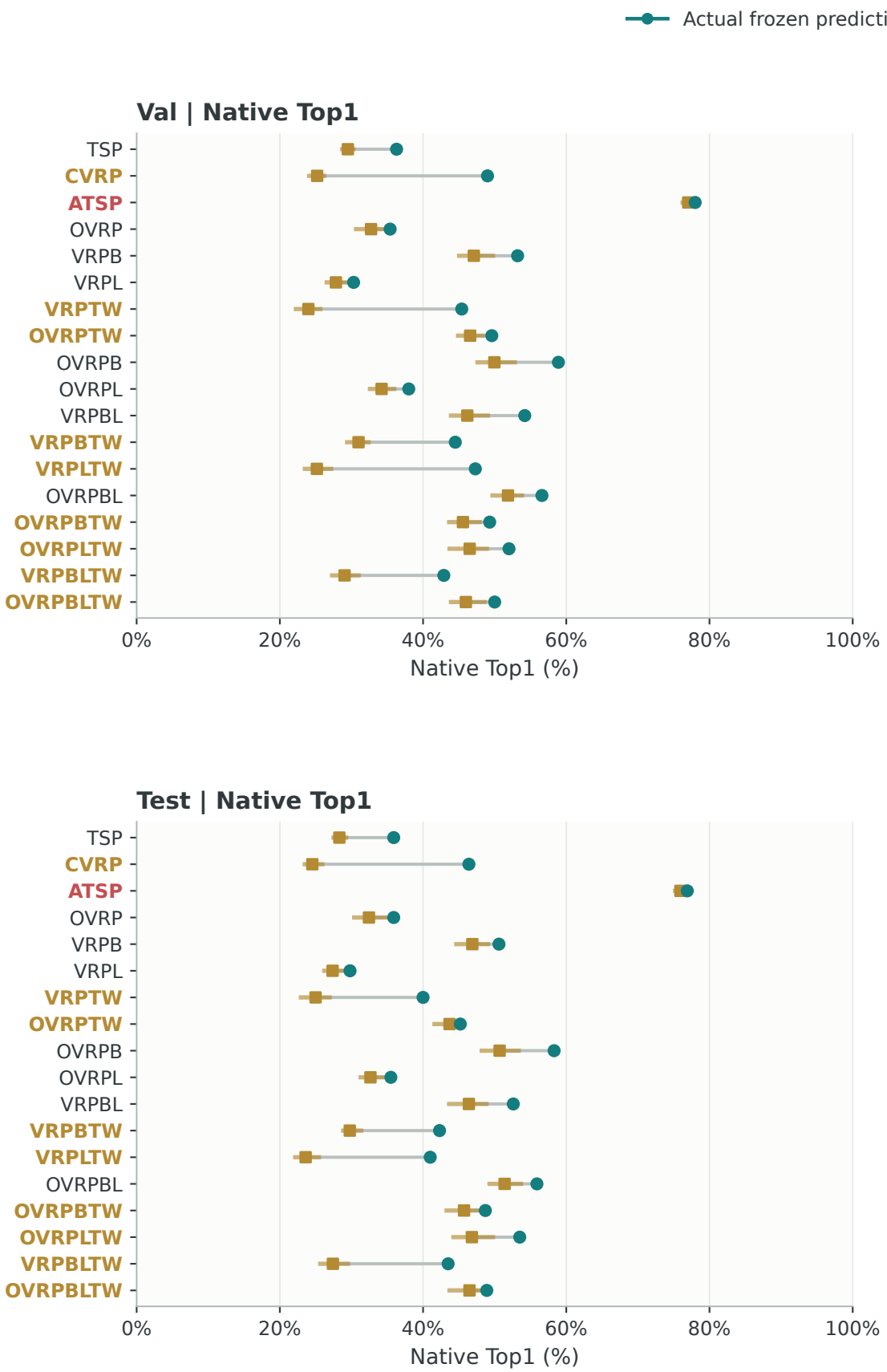
ALL18 = equal weight per task; frozen evaluations only; versions are not independent repetitions.

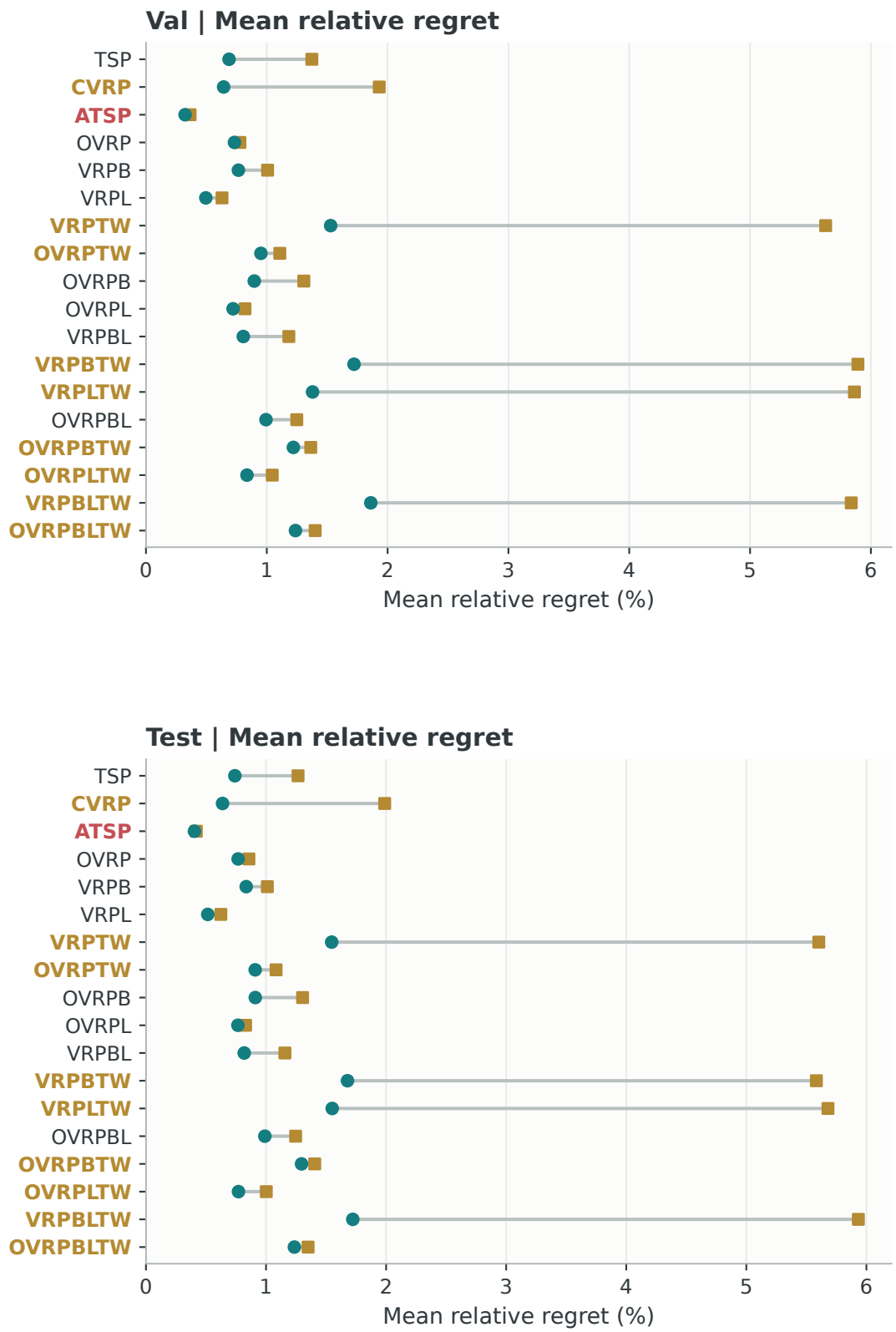


Regret = mean[100 x (selected cost / instance-best cost - 1)]. Missing validation points remain gaps. No four-dataset aggregate and no fitted performance ceiling are shown.

R70B instance signal beyond problem and exact size

Gold task labels: CVRP / time-window variants. Red: ATSP. Each row is one task; no cross-task pooling.

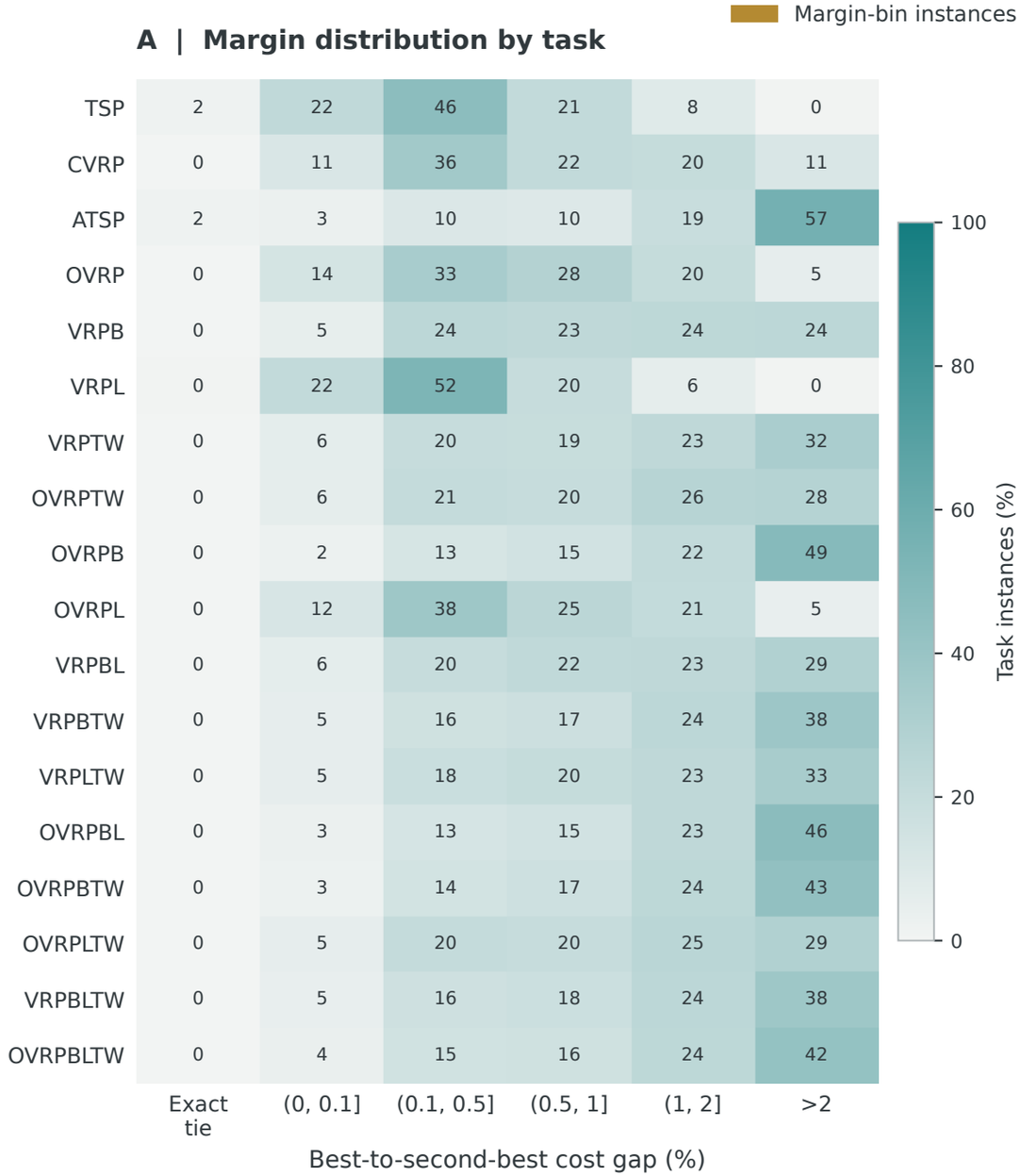


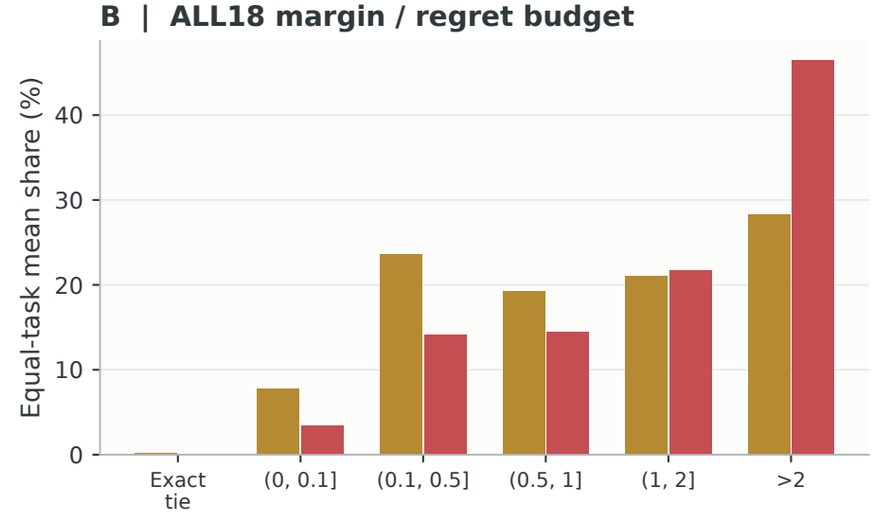


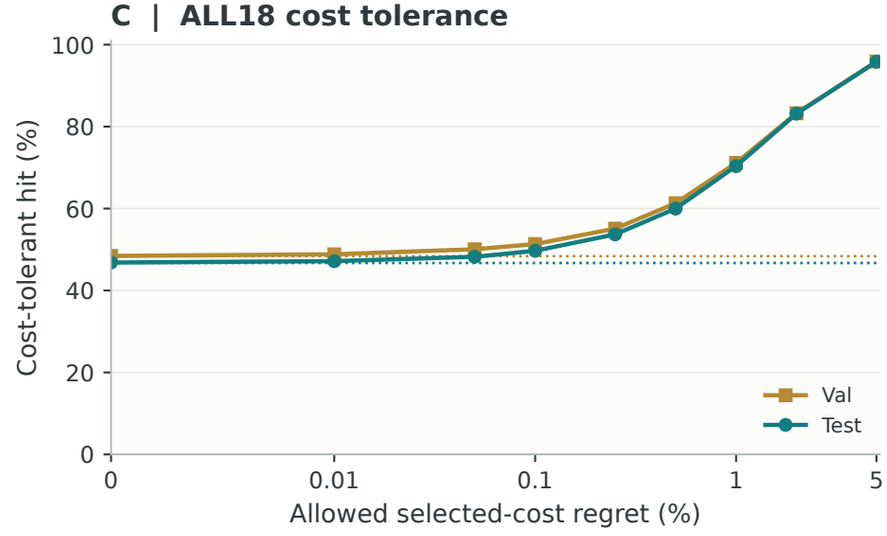
Permutation preserves task, exact true size and pick frequencies. Singleton size groups stay unchanged. Top1 whiskers: 2.5-97.5% of permutation draws, not a sampling confidence interval; full-split estimates.

Winner-runner margins and cost tolerance answer different questions

R70B | Left and upper right: test. Lower right: validation / test. Native Top1 is not a cost-tolerant hit.



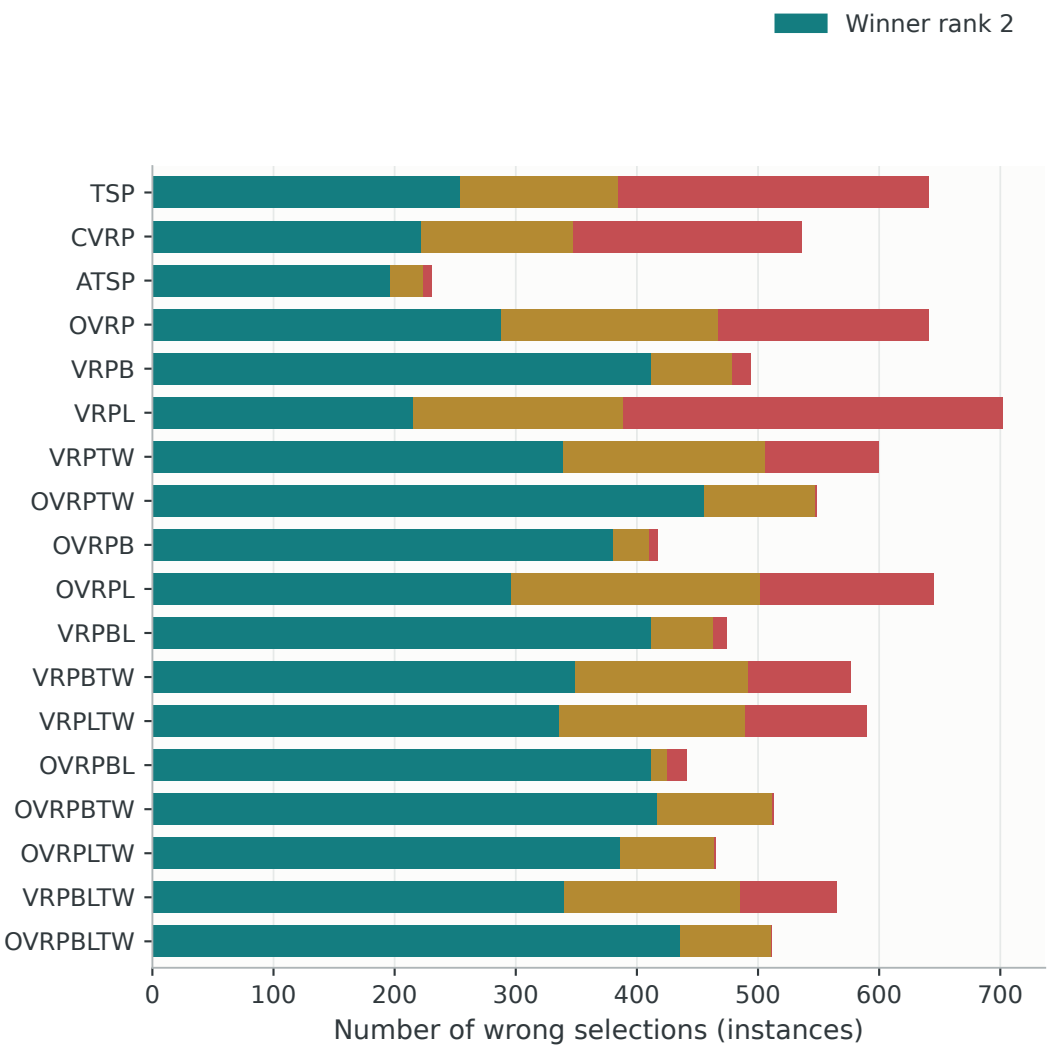


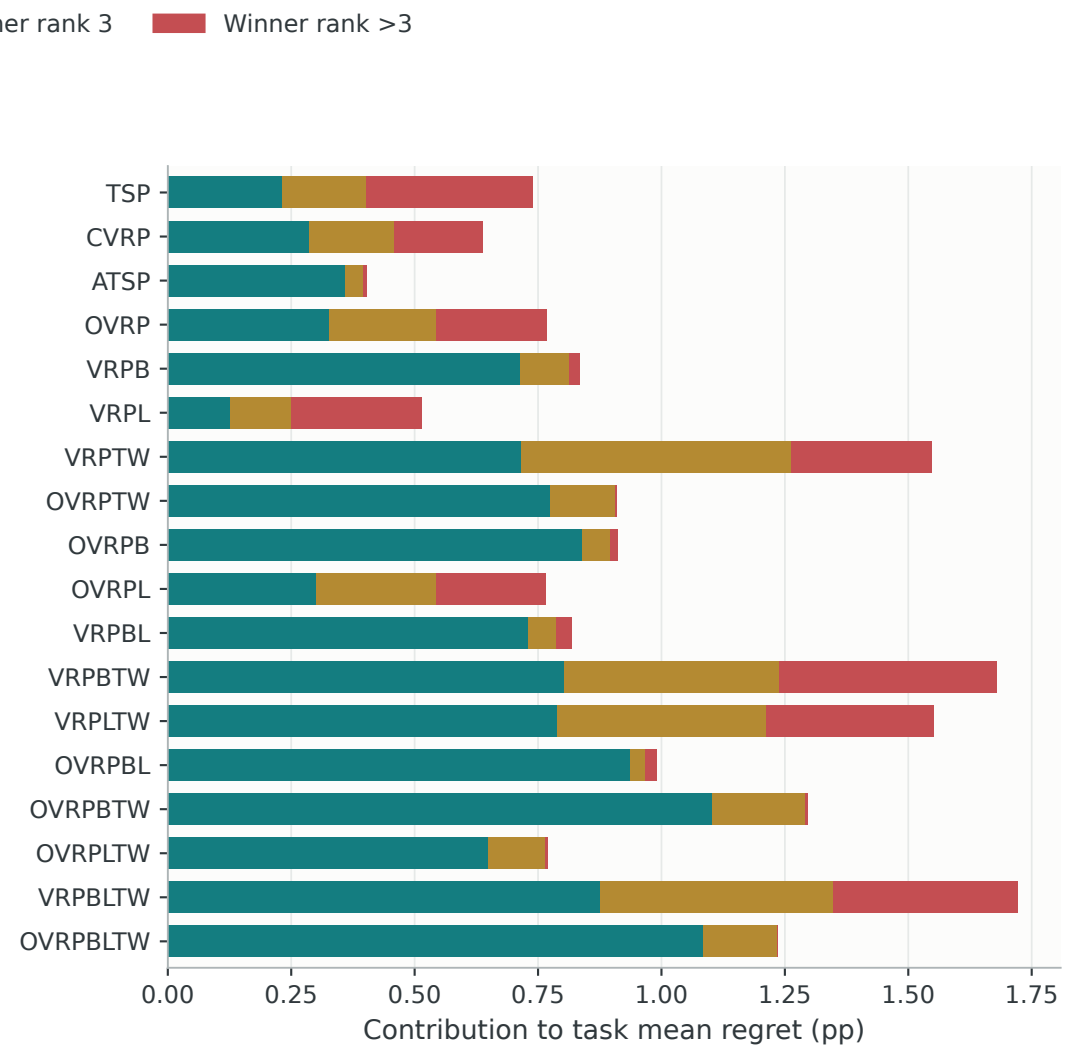


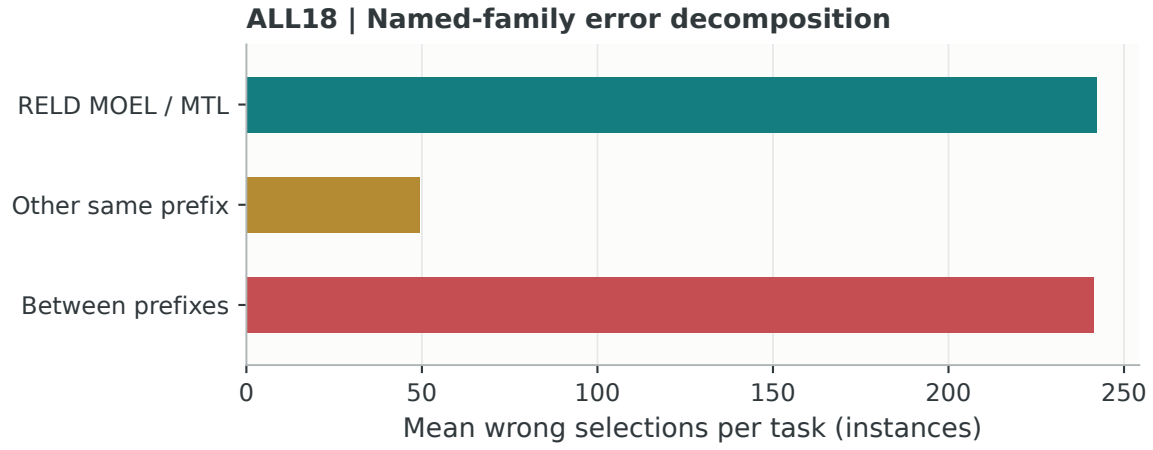
Margin is 100 x (second-best / best - 1); selected-cost regret may exceed this margin. Panel B averages per-task shares, not raw cross-task cost. Dotted lines in C: native Top1; zero tolerance permits exact-cost ties.

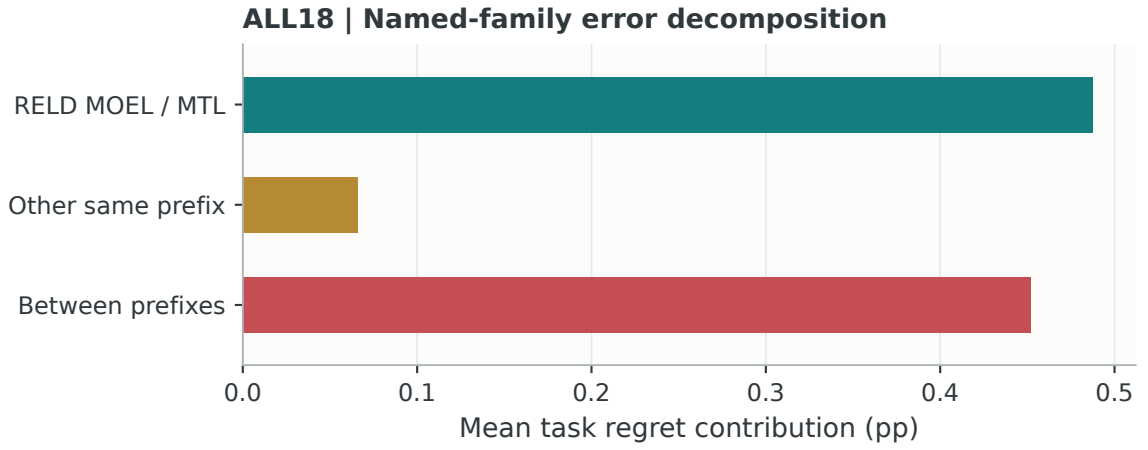
R70B errors: winner rank and named-family cost budgets

Test | Winner ranks refer to the frozen model score order. Cost contributions retain the full task denominator.





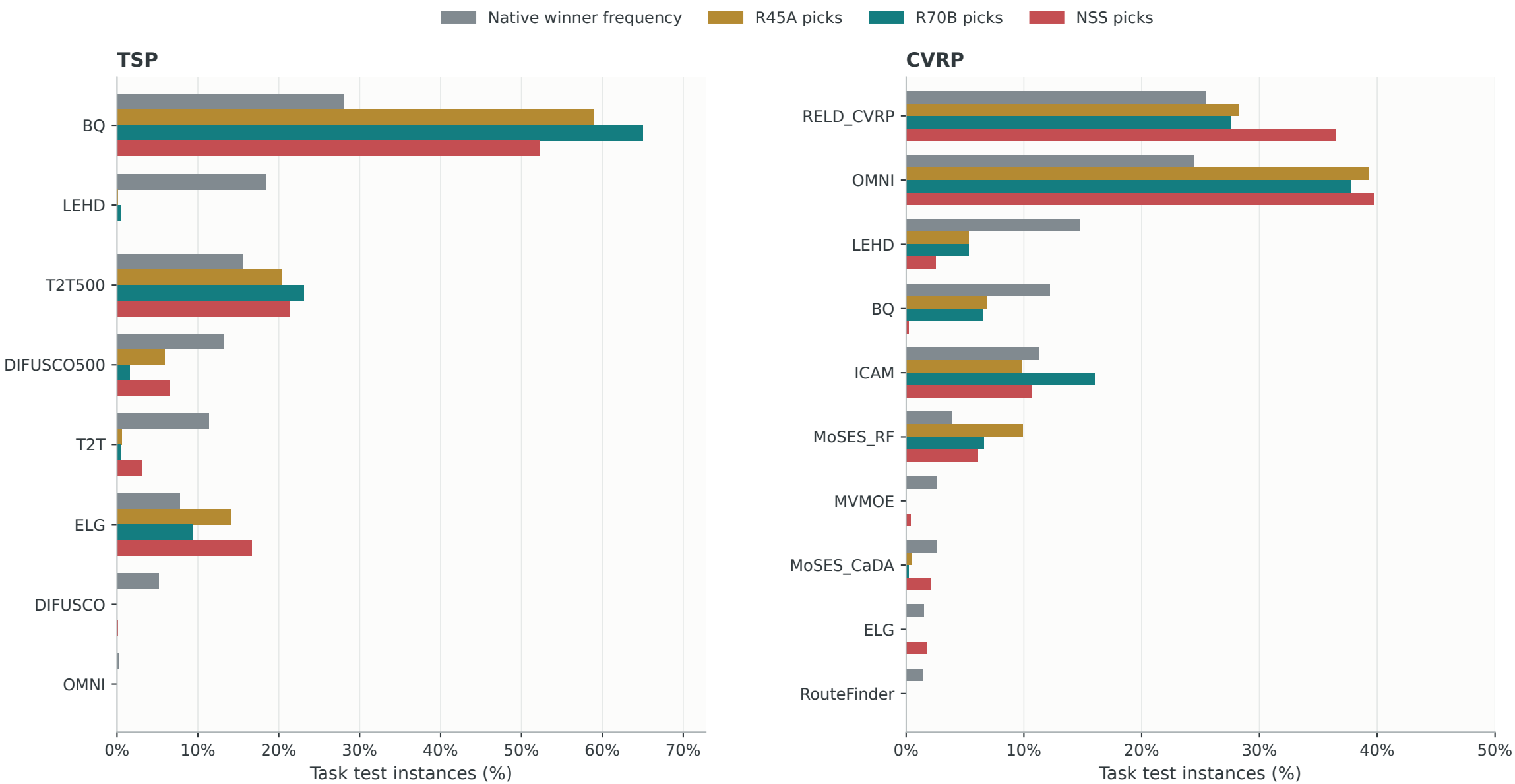




Each task uses its own test denominator. Right: category regret sum / all task instances. Stack totals equal task mean regret, not wrong-only regret; pp = percentage points. Lower panels: equal-task means, complete ALL18 only. Families are descriptive RELD, MoSES, DIFUSCO and T2T name prefixes, not learned architectural groups.

Winner frequencies versus solver picks on TSP and CVRP

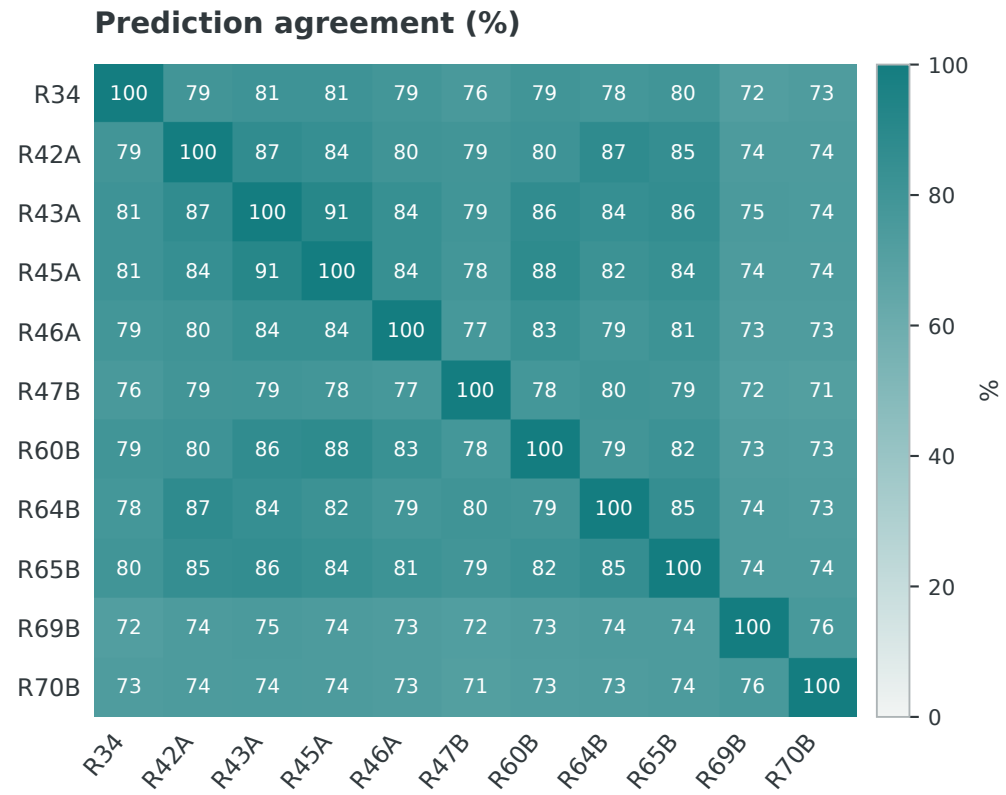
Test | NSS, when present, is the local frozen checkpoint export, not a reconstructed paper result.

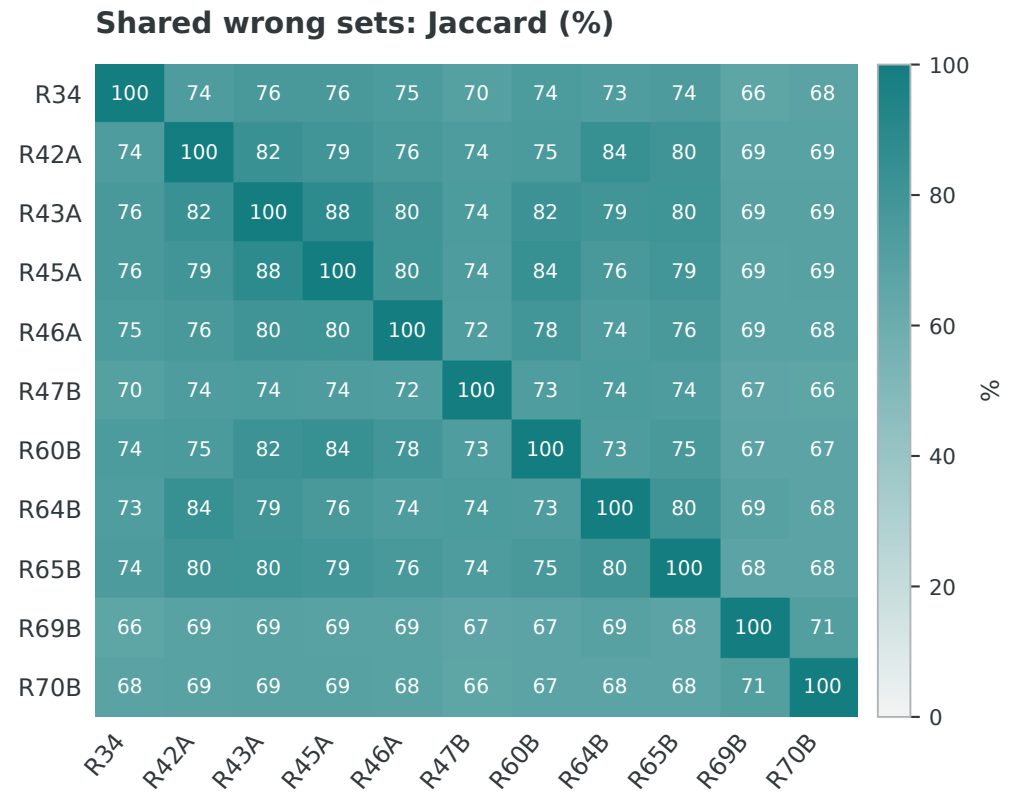


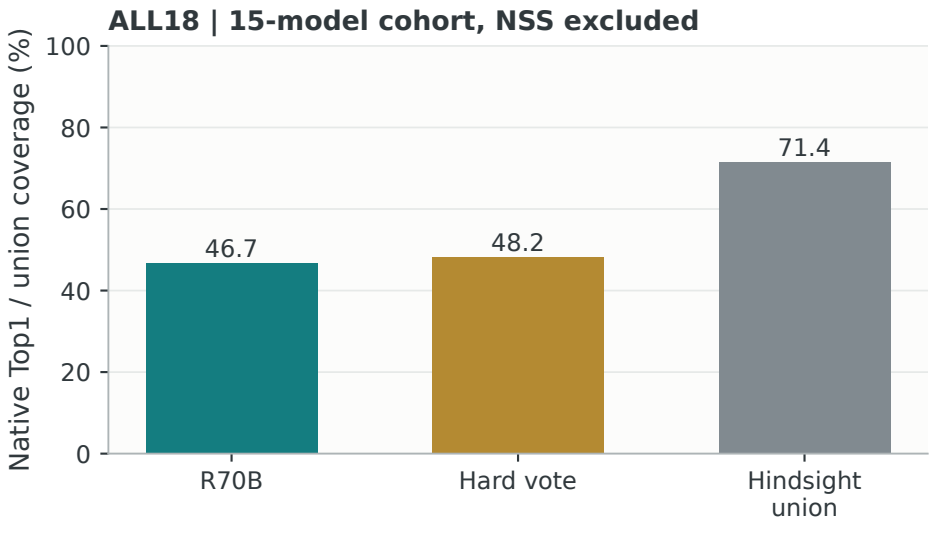
Solver rows are ordered by test winner frequency for display only, not fitted into a selector. A matching marginal pick distribution does not imply correct instance-level assignments. Missing NSS exports are omitted.

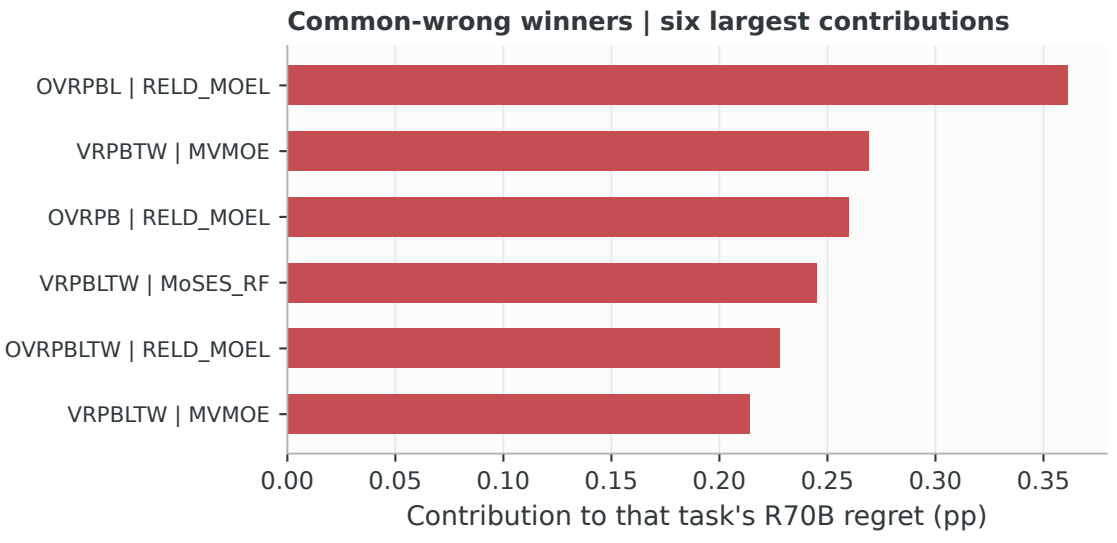
Frozen models agree on predictions and share error sets

Test | Every off-diagonal entry is an equal-task mean over all 18 problems; NSS is not a full-18 model.





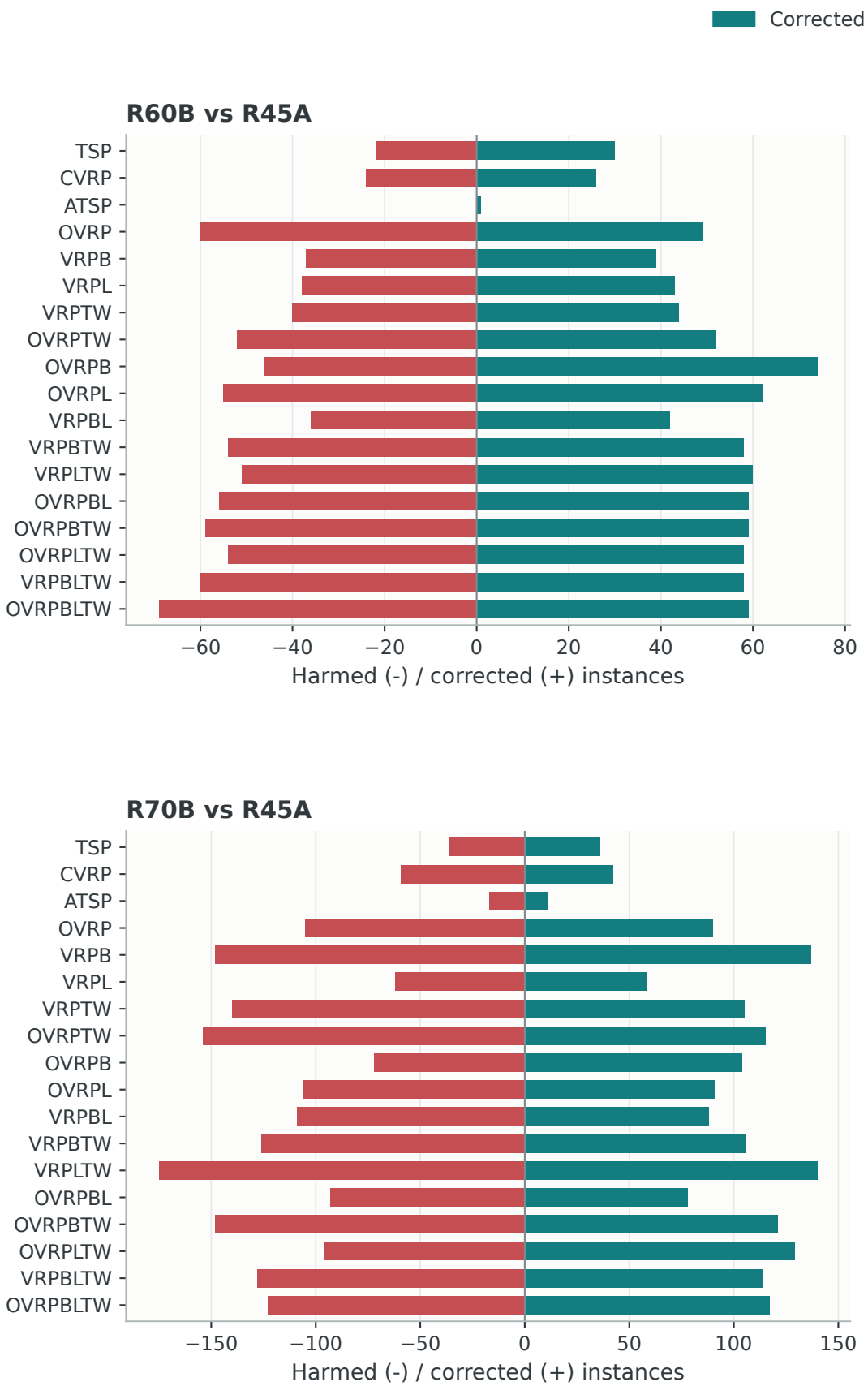




Error Jaccard = wrong for both / wrong for either, within each task before averaging; not joint error rate. Blank matrix cells denote incomplete task pairs. Bottom cohort excludes NSS and can exceed the matrix subset. Union is hindsight coverage, not a deployable selector or a ceiling. Hard vote has unmeasured multi-model overhead; right rows are test-sorted diagnostic task/solver pairs, not a pooled macro.

Changes from R45A: corrected, harmed, and net regret

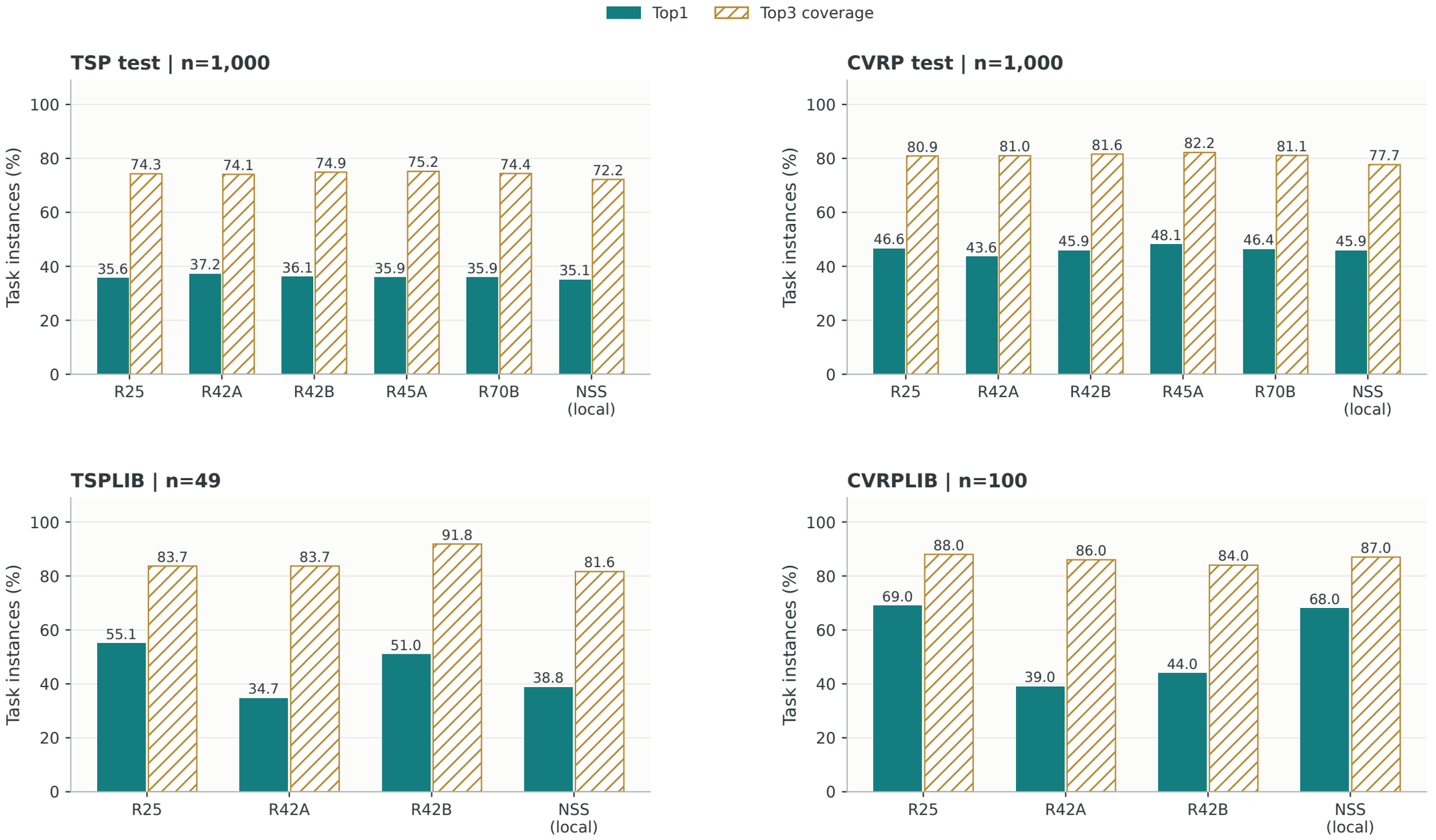
Test | Same instances, frozen versions. Net regret = newer minus R45A; negative is cheaper.



Corrected: R45A wrong, newer right. Harmed: R45A right, newer wrong. Net regret also includes cost changes when both are wrong. Whiskers are descriptive 95% paired-instance bootstrap ranges; one seed, repeated test exposure, no confirmatory significance claim.

Four datasets: native Top1 and winner coverage in Top3

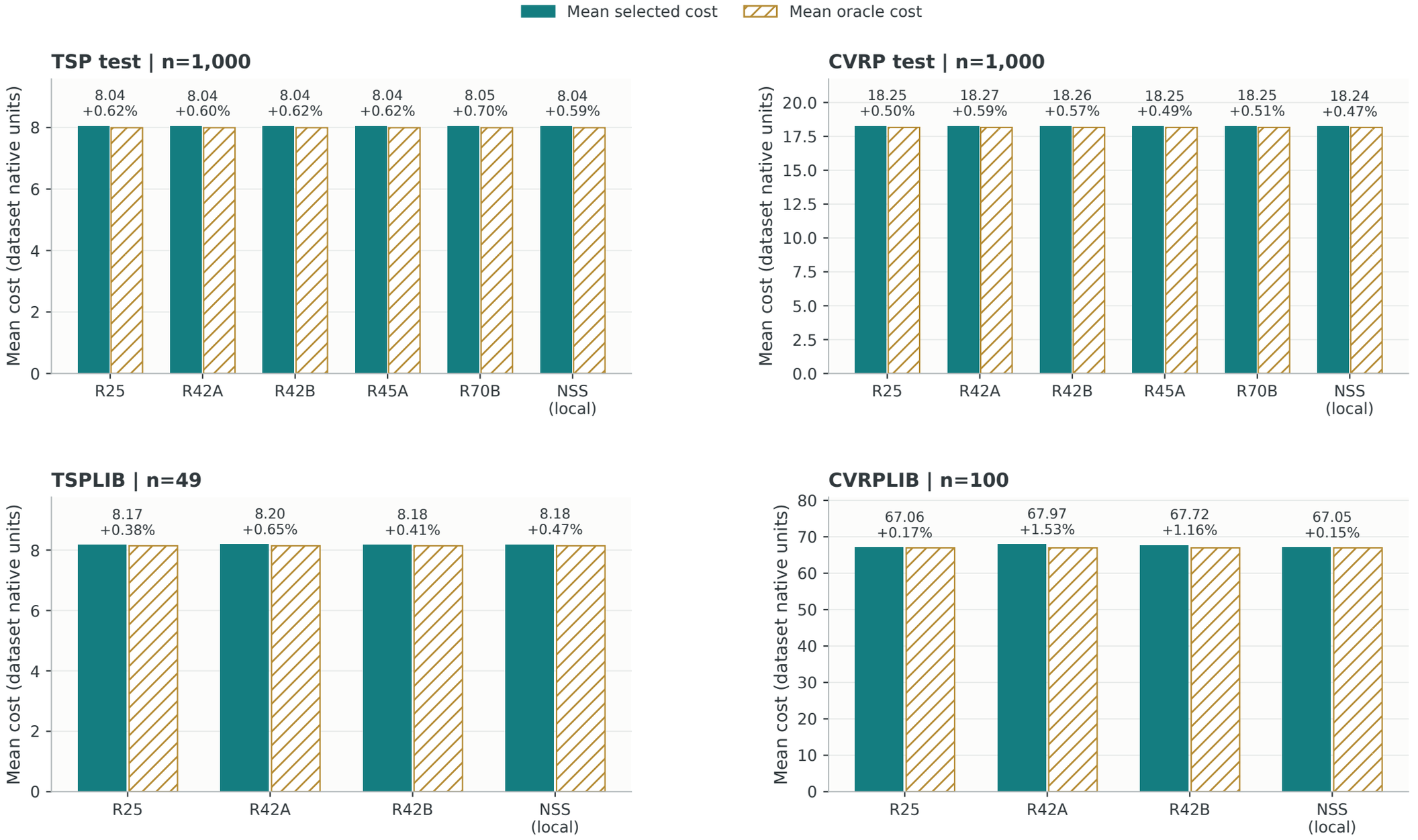
Each dataset stands alone. NSS = local frozen checkpoint export, when available. No four-dataset ALL aggregate.



Top3 is winner-in-shortlist coverage, not single-choice accuracy or a free three-solver result. No original-paper NSS Top3 is inferred. R70B is omitted from LIB unless actual LIB results exist; unavailable model/dataset cells are not filled.

Four datasets: selected cost relative to the instance oracle

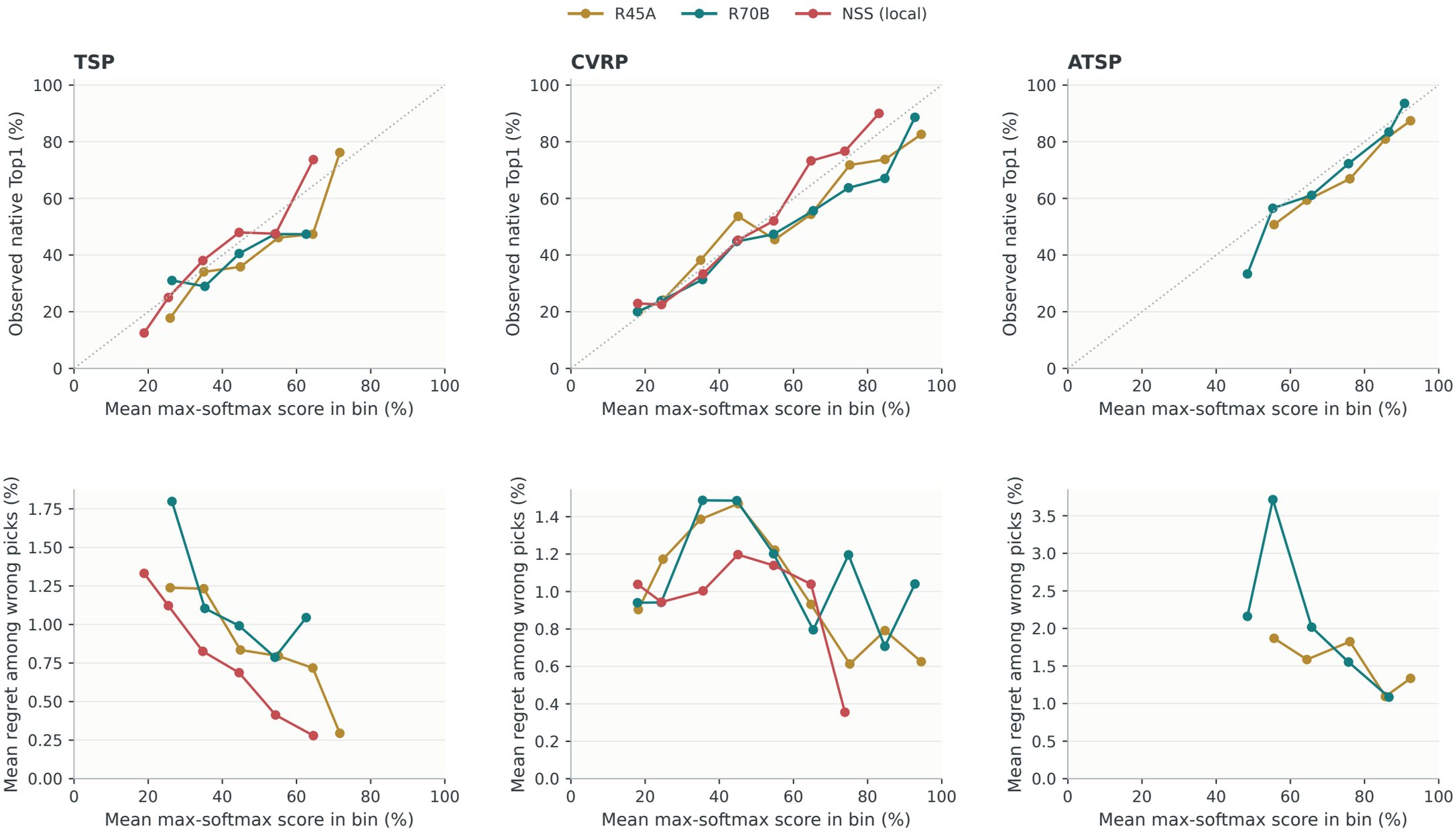
Each dataset stands alone. NSS = local frozen checkpoint export, when available. No four-dataset ALL aggregate.



Annotations: raw mean cost and 100 x (mean selected / mean oracle - 1), a ratio of means, NOT mean instance regret. Cost scales differ across datasets; no pooled raw-cost comparison. R70B is omitted from LIB unless actual LIB results exist; unavailable model/dataset cells are not filled.

Score confidence, empirical correctness, and cost of wrong picks

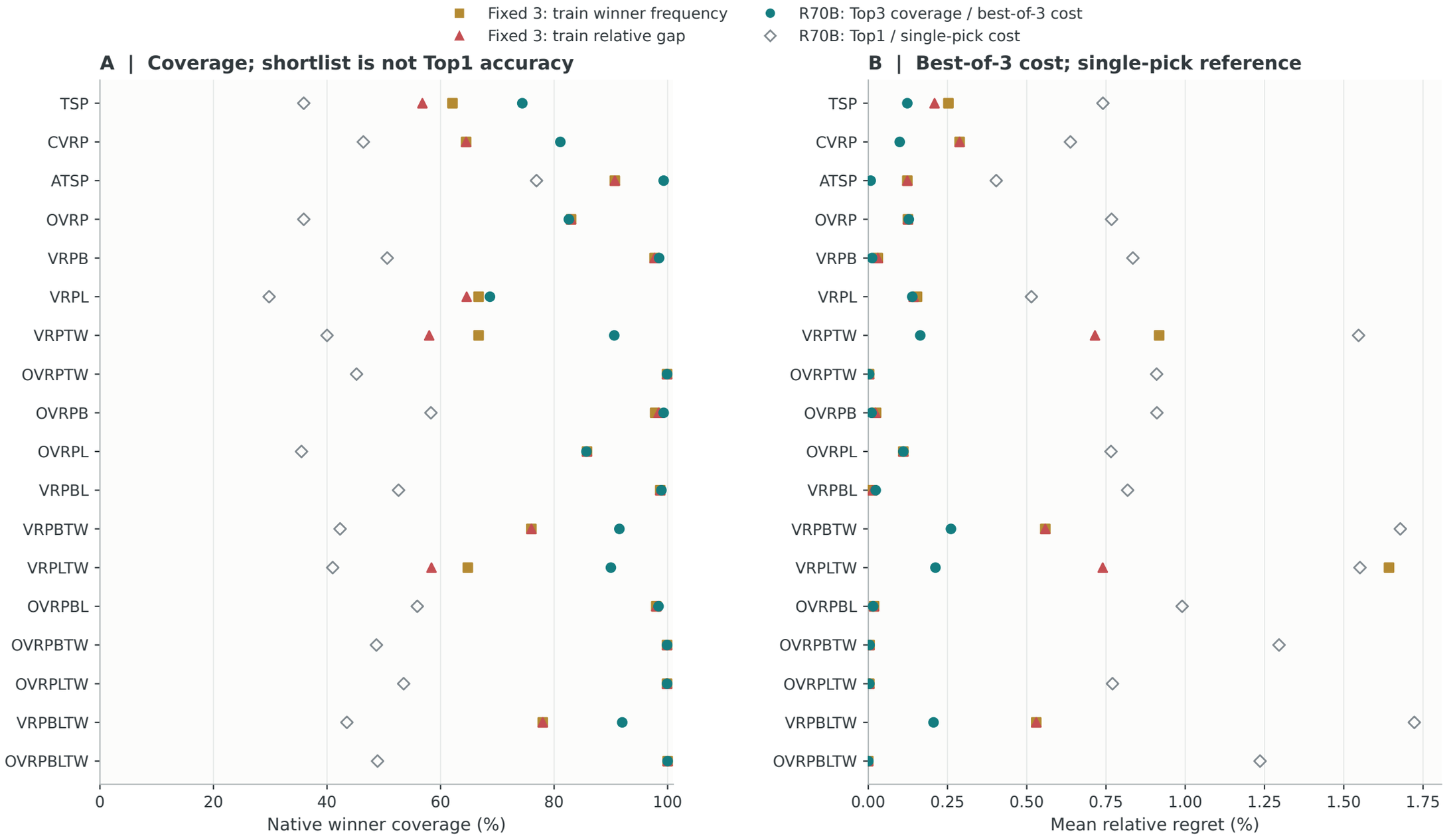
Test | Max-softmax scores are descriptive rankings, not validated or calibrated probabilities. R25 is excluded.



Ten fixed score bins; top row requires >=10 instances, bottom also >=5 wrong picks. Wrong-only costs are calculated directly from existing prediction NPZs, not inferred by dividing bin-average regret by error rate. Dotted diagonal is a visual reference, not a calibration claim.

Train-fixed Top3 portfolios versus adaptive R70B coverage

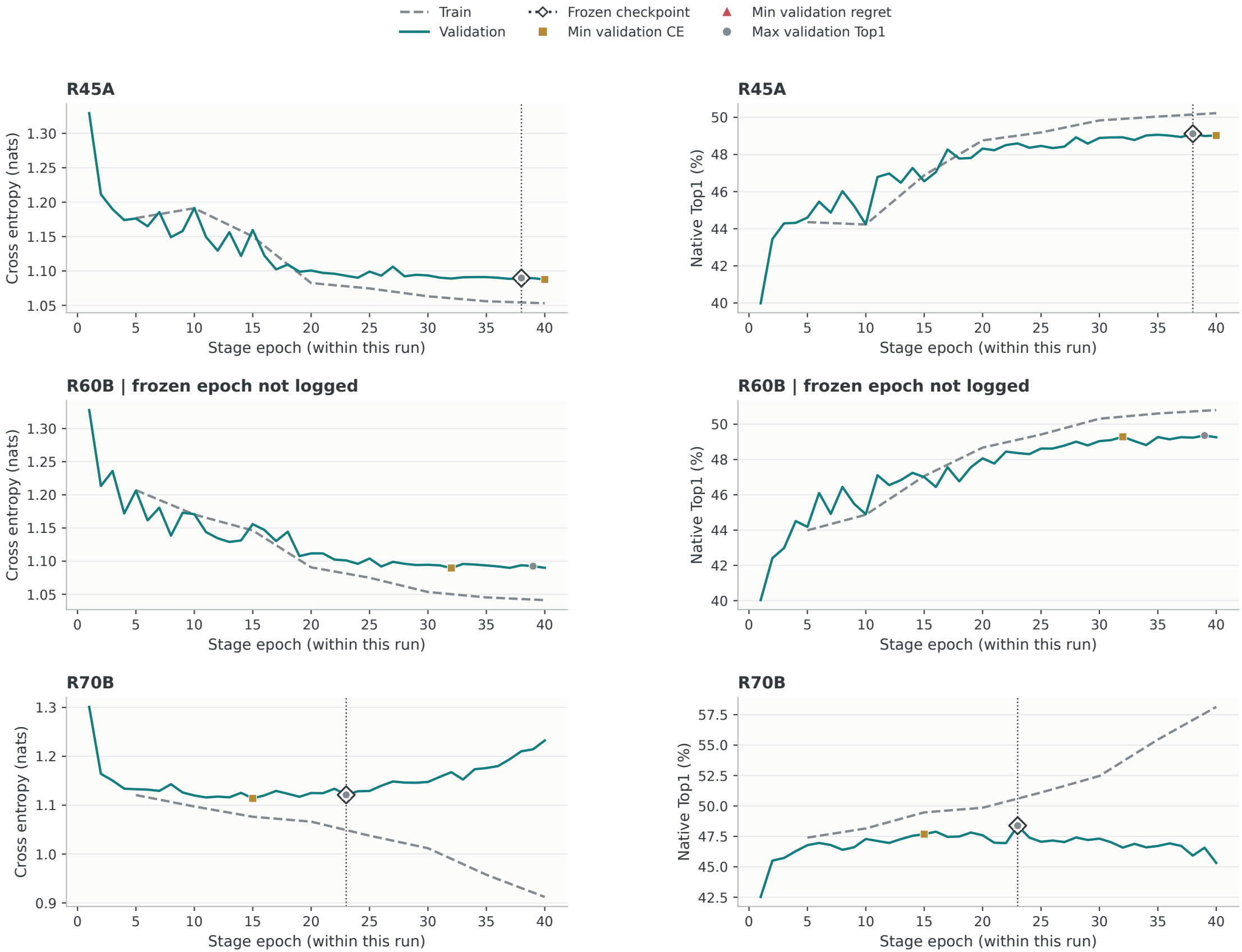
Test | All fixed members were chosen on train only. Three-solver oracle cost is a multi-execution diagnostic.



Fixed and adaptive Top3 costs are the minimum actual cost among three candidates, requiring three solver executions. Open diamonds are the single-pick reference, not the same execution budget. Solver runtime and selector overhead are not compared.

Training histories and validation-only checkpoint diagnostics

ALL18 = equal task means from logged task metrics. Retrospective validation optima do not change the frozen checkpoint.

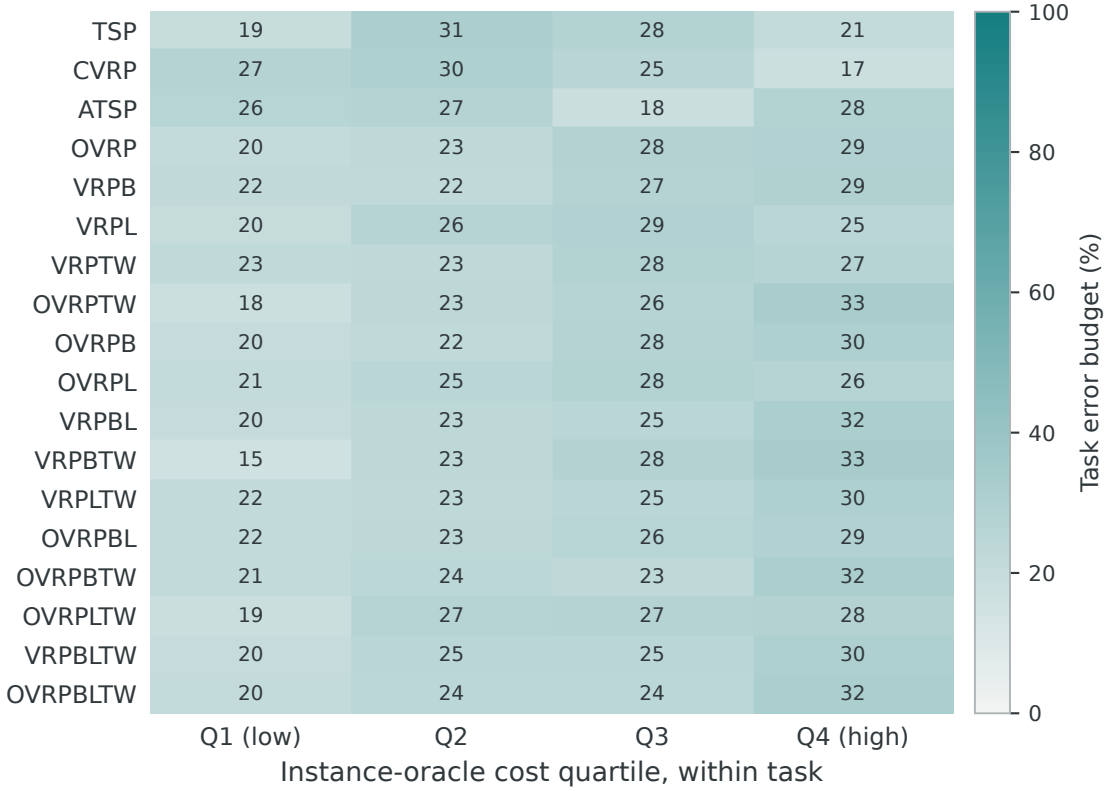


Only training and validation histories appear. Frozen markers come from the logged selected_epoch, never the best test result. R60B has no exported selected_epoch and is not guessed. CE comparisons are within a run, not a claim that distinct losses have identical score semantics.

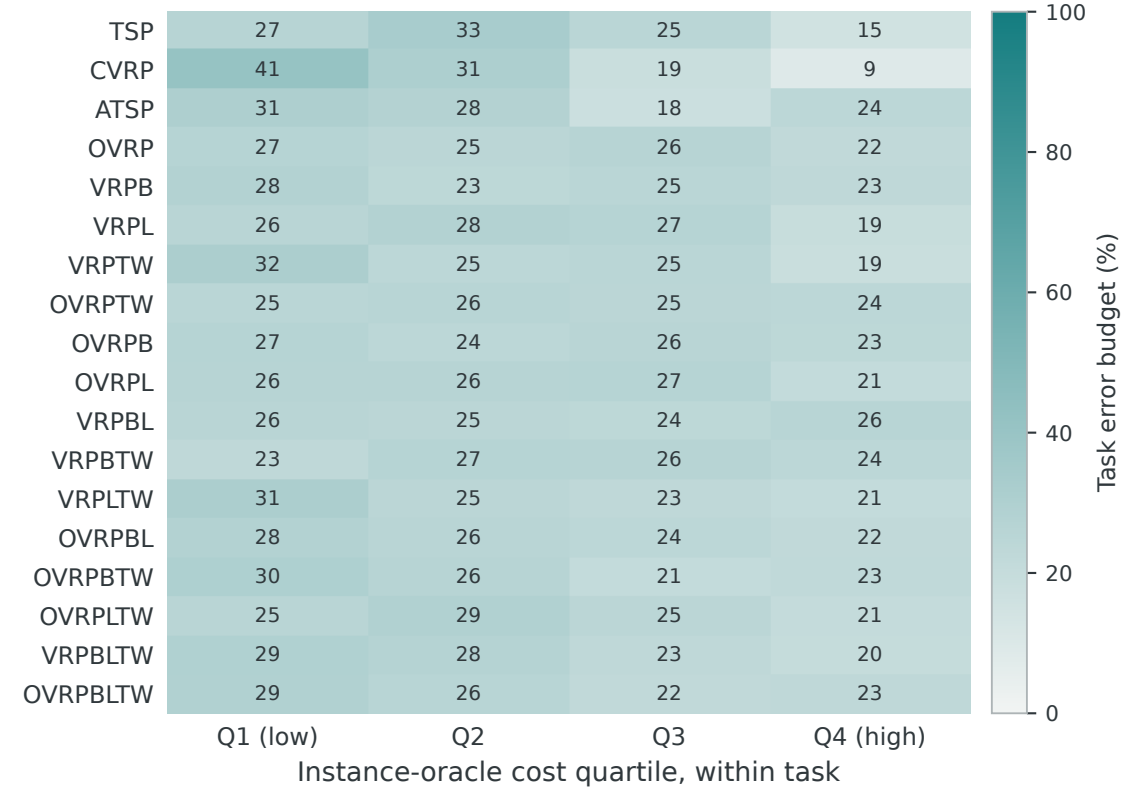
Cost scale and solver rarity explain different parts of the error budget

Test | Upper: R70B, all 18 tasks. Lower: TSP / CVRP solver properties, independent of the selector.

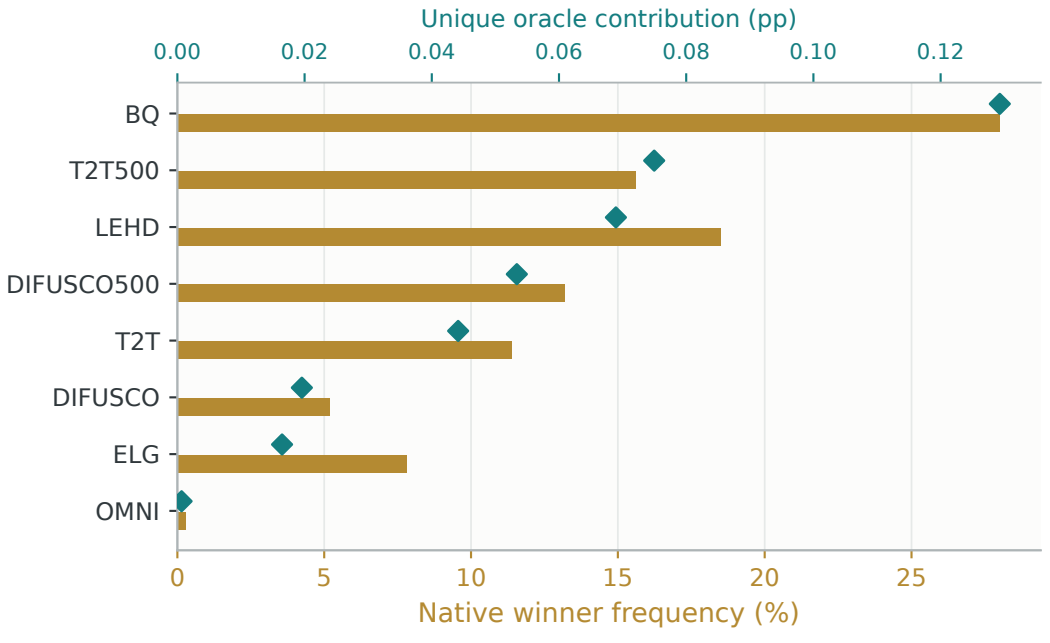
A | Absolute excess-cost share



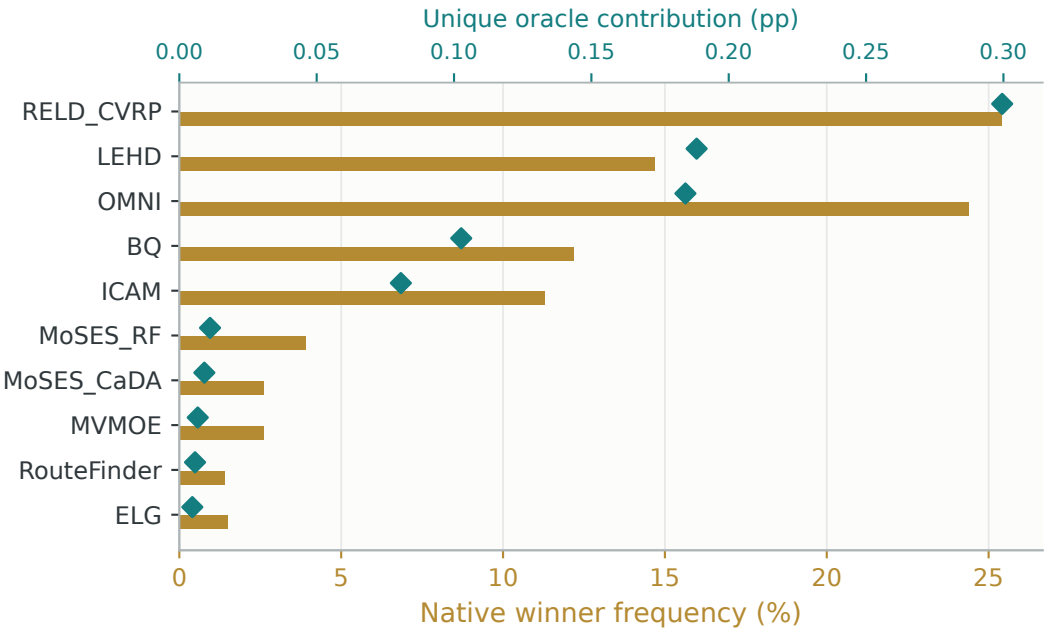
B | Relative-regret share



TSP: frequency vs unique cost contribution



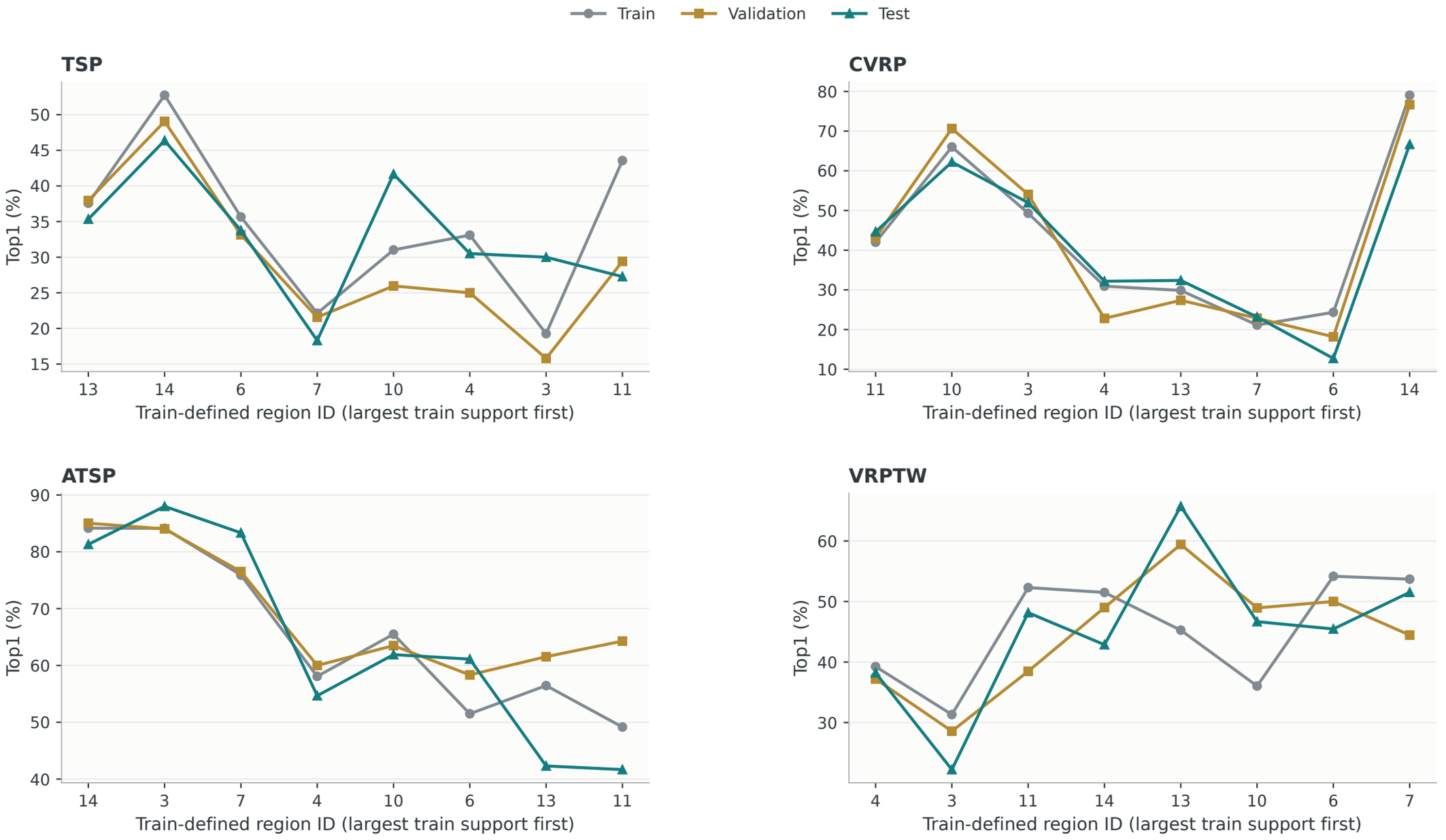
CVRP: frequency vs unique cost contribution



Quartiles are defined on held-out oracle costs for description only. Absolute cost and instance-relative regret use different weights. Unique contribution = mean[max(0, (best cost without solver - solver cost) / oracle) x 100]; ties contribute zero. Bottom axes have different units, not aligned magnitudes.

Train-defined advantage regions: held-out transfer

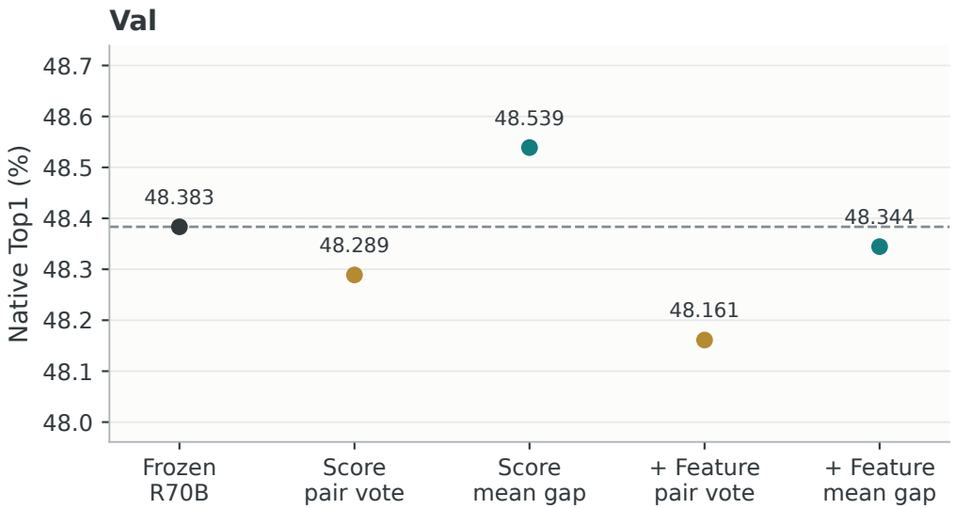
Regions must be fitted on train only. Observed holdout metrics are not a new trained selector.

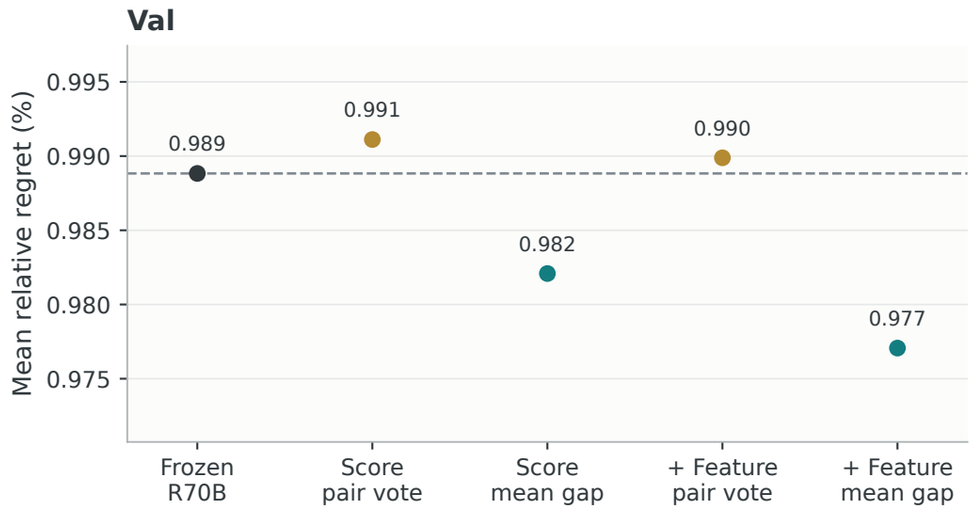


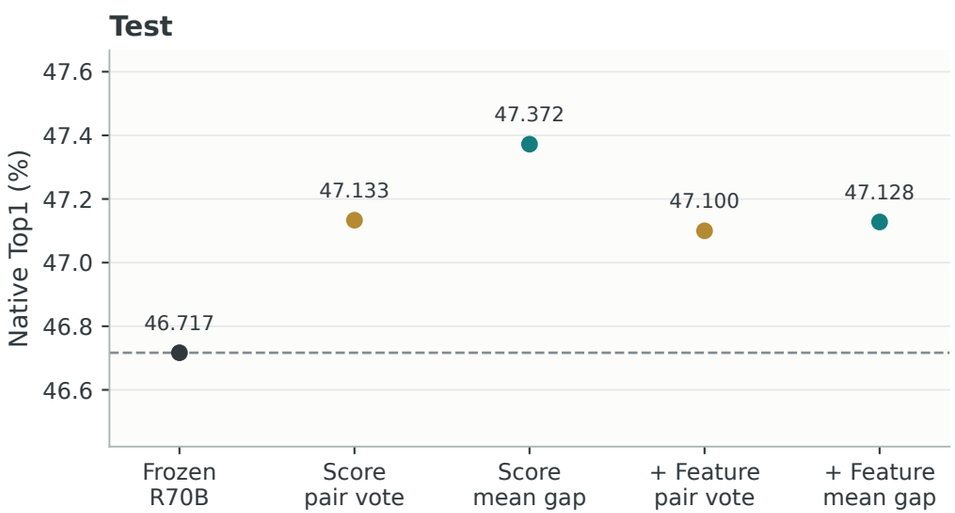
Up to eight regions per task, chosen by train support only, with held-out gaps retained. This plot is conditional on the exported region definition and metric; absence of export is recorded as skipped, not treated as absence of signal.

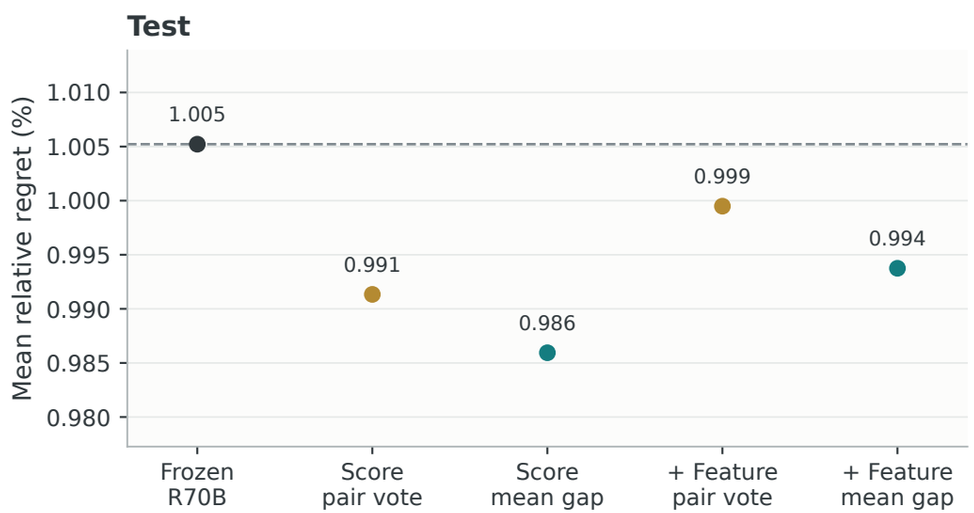
Score-conditioned decisions recover a little; added features are mixed

Train-only conditional tables; same 18 tasks, no neural training, no test-selected rule.





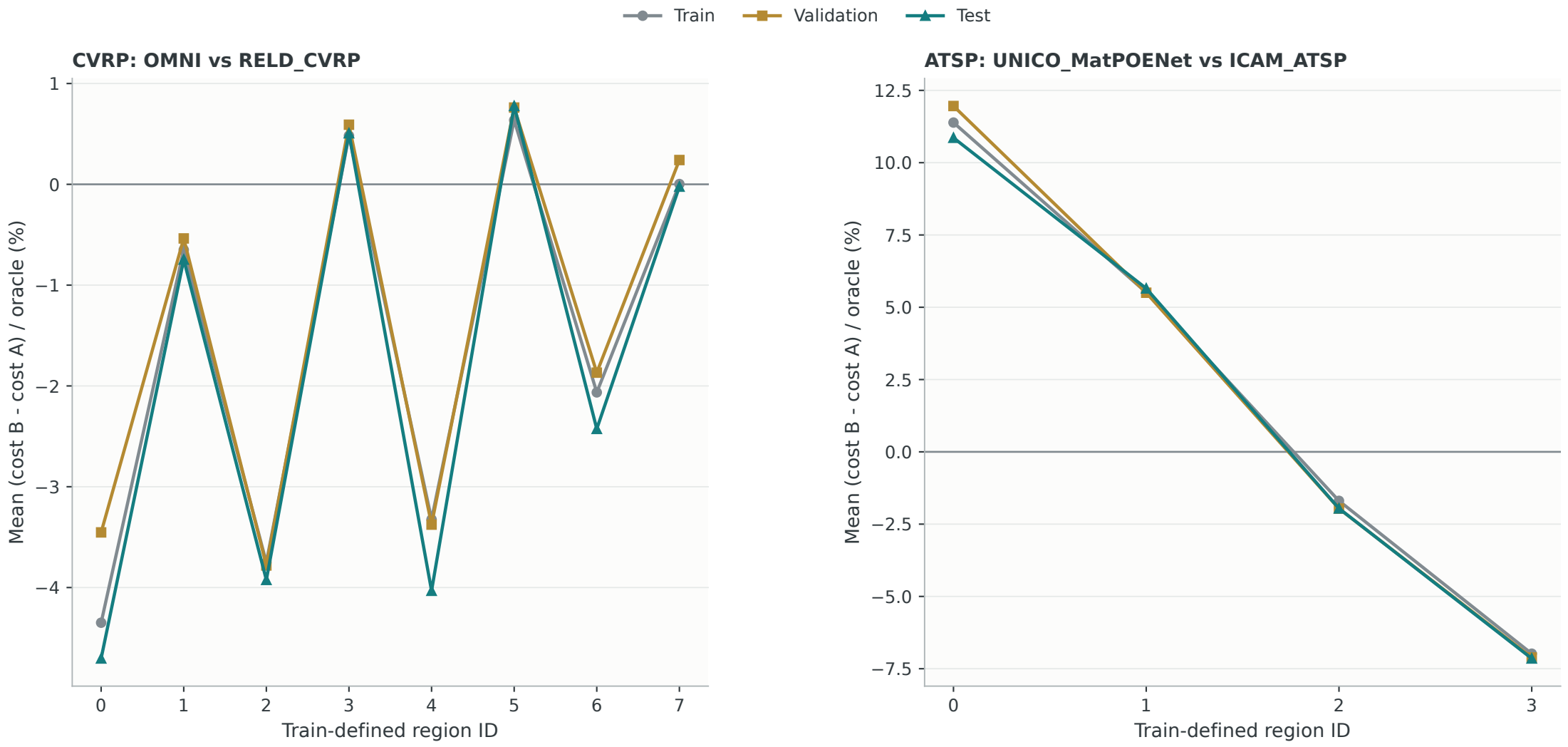




Score cells preserve ordered Top2, train-size quartile and margin tercile. Score/feature tables shrink to the same size/pair parent; a constant feature leaves decisions unchanged. Minimum support 40, pseudocount 50. Test feature-cell coverage 86.3%. Pair vote counts pairwise cost wins, not native full-pool winners. No test rule selection.

Concrete solver preferences transfer across observed instance regions

Positive values favor the first named method; regions and boundaries are fixed using train only.



CVRP IDs 0/1, 2/3, 4/5, 6/7 are size quartiles crossed with low/high normalized demand (train median 0.0468134). ATSP IDs 0-3 are train-size quartiles. All regions shown, not only significant ones. Means do not imply reliable per-instance prediction; these are historical descriptive transfer checks.